

Holistic Cosmology

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The Life of the Cosmos. LEE SMOLIN. Oxford University Press, New York, 1997. viii, 358 pp., illus. \$30. ISBN 0-19-510837-x. Published in the UK by Weidenfeld and Nicolson; £20, ISBN 0-297-81727-2.

The universe is a self-organizing entity. There we have the essence of Lee Smolin's admittedly speculative explanation of everything, from the laws of physics to the origin of intelligent life and even an alternative to God Himself. *The Life of the Cosmos* is an eloquent exposition of a theory that combines cosmology, physics, and philosophy in an attempt to “sketch out a vision” of what our picture of nature will be when the sought-after unification of quantum theory and relativity theory has been achieved. The aims are profound, the vistas outlined are vast, and the discussion is highly creative. What could be more challenging than to construct a Theory of Everything?

Smolin posits that black holes provide the underlying solution to how our universe came to be as it is. The argument proceeds as follows. Every time a black hole forms, a new universe is spun off deep within. Fine so far; this we believe as disciples of Stephen Hawking. Within these universes, the constants of nature suffer small but random changes every time a black hole forms and a new universe develops. The most probable universe emerges from this evolutionary sequence with the maximal number of evolutionary pathways, and hence black holes. We are at the peak of a grand cosmic summit of natural selection of the laws of physics. And—here is the beauty of the argument—such a course of events is inevitable: we are the outcome. Not a bad postulate if it is testable, which Smolin argues to be the case. For example, one need only search for the black holes.

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Sadly, however, the challenge Smolin poses fails at almost every encounter with astronomical reality. Our universe is far—by about four orders of magnitude, or a factor of 10,000—from being optimally loaded with black holes. Most cosmologists are convinced that enhancing the amplitude of primordial density fluctuations would enhance the black hole fraction. We are very far from saturation. Smolin's response to this criticism is an appeal to self-regulation of the inflaton, the field responsible for inflation, a concept that is beyond any present physics of which I am aware. He argues that small changes in fundamental parameters won't do much. That is simply untrue. Tilt the spectral index of primordial fluctuations blueward by 10 percent and one would fill the early universe with col-

lapsed objects, possibly destined to make black holes. Even worse, imagine adding the tiniest admixture of black holes early on. Perhaps during an early phase transition some rare horizon volume received a fatal compression that pushed it over the precipice of black hole formation. The primordial black holes conserve their mass as the dominant relativistic energy density of the universe redshifts away. The result: one in a billion, even one in a trillion, is all it might take in terms of large primordial horizon-scale overdensities to result in a universe that today is vastly overloaded with black holes compared to what we see around us.

And dare I mention star formation? Smolin exposes the hazards for a quantum cosmologist of dabbling in murky waters that are governed more by astronomical common sense than by fundamental physics. He asserts that carbon is needed to form stars. Think of the obvious counterexample, us—or, more specifically, our antecedents, namely the first stars, or for that matter the metal-poor halo stars, which are as close to being carbon-free as makes a shred of difference to gas cooling. Nor need this be a puzzle: hydrogen cools as efficiently as does carbon, albeit at a higher temperature. But there is no limit to how small or large a lump of hydrogen one could fabricate, short of making snowballs of solid hydrogen, and stellar masses are well within the range of possibility.

And then there is feedback, a continuing theme. Feedback is crucial to self-organization. But would it not be advisable to know the sign of the feedback? Negative and positive feedback could differentiate ants from whales, or a Big Crunch from a Big Bang. It may suffice to give an example of the alarmingly large astrophysical uncertainties: reduce the number of supernovae, and one diminishes feedback. In this case the gas collapses with higher efficiency, making more stars and more black holes. This is quite the opposite of what Smolin tells us: no supernovae, no carbon, no stars, no black holes. Which is the truth and which is the fallacy? Nobody knows.

Perhaps these awkwardnesses offer a cautionary tale for physicists with cosmological inclinations, indicating the pitfalls of dabbling in other disciplines before constructing a more solid edifice. Of course, metaphysical cosmology is a well-trodden path, and Smolin is by no means the first to pursue it. No doubt this explains why his tome has received some surprisingly resounding endorsements, and I must admit, finally, that the book is worth reading, if only to see how a brave pioneering spirit can conjure up impressive mirages of the ultimate theory.

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